2023
VCE
Specialist Mathematics
Year 12
Trial Examination 1

Detailed Answers

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Question 1

$z^4 + 16 = 0$

$z^4 = -16$

$z^4 = 16\operatorname{cis}(\pi + 2k\pi)$ A1

$z = 16^{\frac{1}{4}}\operatorname{cis}\left(\dfrac{\pi + 2k\pi}{4}\right)$

$z = 2\operatorname{cis}\left(\dfrac{\pi(1+2k)}{4}\right)$ let $k = -2, -1, 0, 1$ M1

$k = -2 \quad z = 2\operatorname{cis}\left(-\dfrac{3\pi}{4}\right) = \sqrt{2}(-1-i)$

$k = -1 \quad z = 2\operatorname{cis}\left(-\dfrac{\pi}{4}\right) = \sqrt{2}(1-i)$

A2

$k = 0 \quad z = 2\operatorname{cis}\left(\dfrac{\pi}{4}\right) = \sqrt{2}(1+i)$

$k = 1 \quad z = 2\operatorname{cis}\left(\dfrac{3\pi}{4}\right) = \sqrt{2}(-1+i)$

Question 2

$\dfrac{dy}{dx} = \dfrac{x(9+4y^2)}{6}, \quad y(0) = \dfrac{3}{2}$ separating the variables

$\displaystyle\int \frac{1}{9+4y^2}\,dy = \frac{1}{6}\int x\,dx$

$\dfrac{1}{6}\tan^{-1}\left(\dfrac{2y}{3}\right) = \dfrac{x^2}{12} + c$ A1

To find c, when $x = 0,\ y = \dfrac{3}{2},\ \dfrac{1}{6}\tan^{-1}(1) = \dfrac{1}{6} \times \dfrac{\pi}{4} = 0 + c,\ c = \dfrac{\pi}{24}$ M1

$\dfrac{1}{6}\tan^{-1}\left(\dfrac{2y}{3}\right) = \dfrac{x^2}{12} + \dfrac{\pi}{24}$

$\tan^{-1}\left(\dfrac{2y}{3}\right) = \dfrac{x^2}{2} + \dfrac{\pi}{4}$

$\dfrac{2y}{3} = \tan\left(\dfrac{x^2}{2} + \dfrac{\pi}{4}\right)$

$y = \dfrac{3}{2}\tan\left(\dfrac{x^2}{2} + \dfrac{\pi}{4}\right)$ A1

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Question 3

$$f(x)=\frac{x}{x^2-4}=\frac{x}{(x+2)(x-2)} \quad \text{domain } D=R\setminus\{\pm 2\}$$

i. $$f'(x)=\frac{1(x^2-4)-2x^2}{(x^2-4)^2}=\frac{-(x^2+4)}{(x^2-4)^2}\neq 0$$ so there are no turning points A1

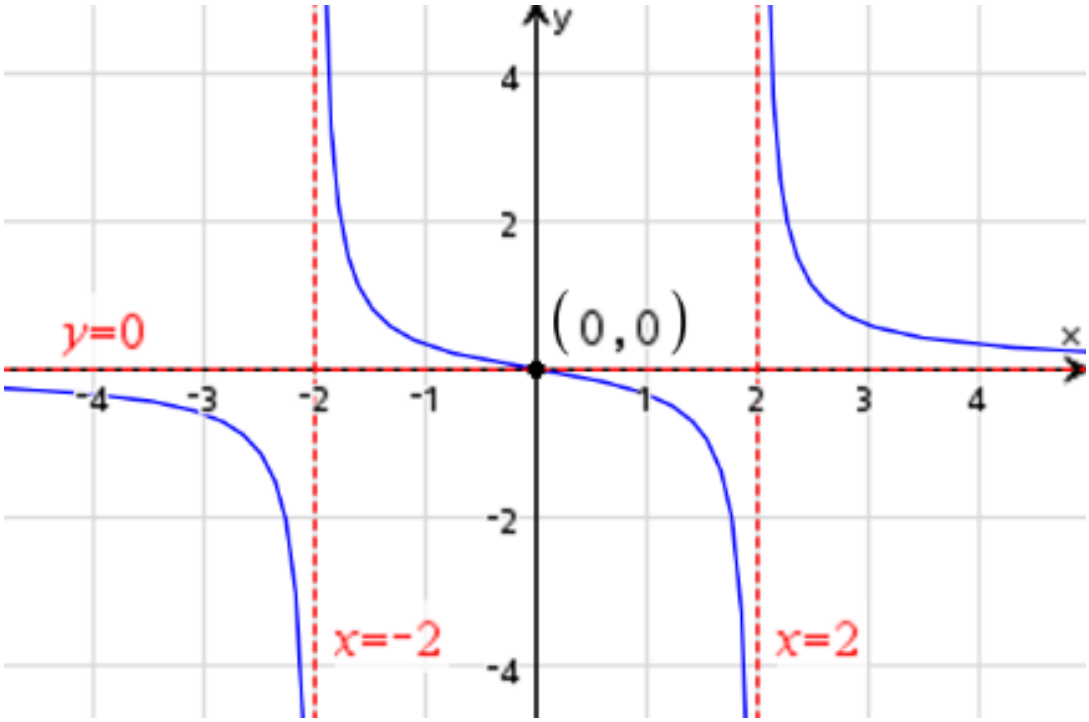
ii. There are vertical asymptotes at $x=\pm 2$ and there is a horizontal asymptote at $y=0$.

The graph crosses the x and y axis at the origin $(0,0)$ and this is the point of inflexion as

$$f''(x)=\frac{2x(x^2+12)}{(x^2+4)^3}=0, \Rightarrow x=0 \text{ and when } -2<x<0,\ y>0 \text{ and when } 0<x<2,\ y<0$$

Also when $x>2,\ y>0$, and when $x<-2,\ y<0$. A1

G1



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Question 4

a. $a=\frac{d}{dx}\left(\frac{1}{2}v^2\right)=-\frac{4}{3}e^{-\frac{2x}{3}}$

$\frac{1}{2}v^2=-\frac{4}{3}\int e^{-\frac{2x}{3}}\,dx=-\frac{4}{3}\times\frac{-3}{2}e^{-\frac{2x}{3}}+k$ M1

$v^2=4e^{-\frac{2x}{3}}+c_1,$

To find c_1, when $x=0,\ v=2,\quad 4=4+c_1,\quad \Rightarrow c_1=0$

$v=2e^{-\frac{x}{3}}$ since $v>0$

b. $v=\frac{dx}{dt}=2e^{-\frac{x}{3}}$ inverting $\frac{dt}{dx}=\frac{1}{2}e^{\frac{x}{3}}$

$t=\frac{1}{2}\int e^{\frac{x}{3}}dx=\frac{3}{2}e^{\frac{x}{3}}+c_2$

To find c_2, when $t=0,\ x=0,\quad 0=\frac{3}{2}+c_2,\quad \Rightarrow c_2=-\frac{3}{2}$ M1

$t=\frac{3}{2}\left(e^{\frac{x}{3}}-1\right)\qquad \frac{2t}{3}=e^{\frac{x}{3}}-1$

$e^{\frac{x}{3}}=\frac{2t}{3}+1=\frac{2t+3}{3}$

$x=3\log_e\left(\frac{2t+3}{3}\right)$ when $t=3$

$x=3\log_e(3)=\log_e(27),\quad s=27$ A1

Question 5

a. $C\overset{d}{=}N(100,12),\qquad T=C_1+C_2+C_3$ total of three independent pieces of corn

$E(T)=3E(C)=3\times100=300$

$\text{Var}(T)=3\text{Var}(C)=3\times12=36\qquad \sigma_T=\sqrt{36}=6$

$T\overset{d}{=}N\left(100,6^2\right)$

$\Pr(291<T<300)=\Pr\left(\frac{291-300}{6}<Z<\frac{300-300}{6}\right)$

$=\Pr(-1.5<Z<0)=\Pr(0<Z<1.5)$ M1

$=\Pr(Z<1.5)-\Pr(Z<0)=\Pr(Z<1.5)-0.5$

$=0.933-0.5$

$=0.433$ A1

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b. $\bar{x}=100,\quad s^2=12,\quad 95\%,\quad z=2$

$$\left(\bar{x}-z\frac{s}{\sqrt{n}},\bar{x}-z\frac{s}{\sqrt{n}}\right)=(99,101)$$

$$2z\frac{s}{\sqrt{n}}=2$$

$$2\sqrt{\frac{12}{n}}=1$$

$$\sqrt{\frac{12}{n}}=\frac{1}{2}$$

$$\frac{n}{12}=4$$

$$n=48$$ A1

Question 6

a. $\frac{dK}{dt}=\frac{K}{5}\left(1-\frac{K}{500}\right)=\frac{K}{5}\left(\frac{500-K}{500}\right)=\frac{K(500-K)}{2500},\quad K(0)=50$

inverting both sides $\frac{dt}{dK}=\frac{2500}{K(500-K)}$ now by partial fractions

$$\frac{2500}{K(500-K)}=\frac{A}{K}+\frac{B}{500-K}=\frac{A(500-K)+BK}{K(500-K)}=\frac{500A+K(B-A)}{K(500-K)}$$ M1

equating $(1)\,B-A=0,\quad (2)\,500A=2500,\quad \Rightarrow A=B=5$

$$t=5\int\left(\frac{1}{K}+\frac{1}{500-K}\right)dK$$

$\frac{t}{5}=\log_e(|K|)-\log_e(|500-K|)+c$ since $50\le K<500$, the moduli are not needed.

$\frac{t}{5}=\log_e\left(\frac{K}{500-K}\right)+c$ using $t=0,\ K=50$ to find c A1

$$0=\log_e\left(\frac{50}{500-50}\right)+c\quad\Rightarrow c=-\log_e\left(\frac{50}{450}\right)=-\log_e\left(\frac{1}{9}\right)=\log_e(9)$$

$$\frac{t}{5}=\log_e\left(\frac{K}{500-K}\right)+\log_e(9)=\log_e\left(\frac{9K}{500-K}\right)$$

$$e^{\frac{t}{5}}=\frac{9K}{500-K},\qquad \frac{500-K}{9K}=e^{-\frac{t}{5}}$$ M1

$$500-K=9Ke^{-\frac{t}{5}},\qquad 500=K+9Ke^{-\frac{t}{5}}=K\left(1+9e^{-\frac{t}{5}}\right)$$

$$K=K(t)=\frac{500}{1+9e^{-\frac{t}{5}}}$$ A1

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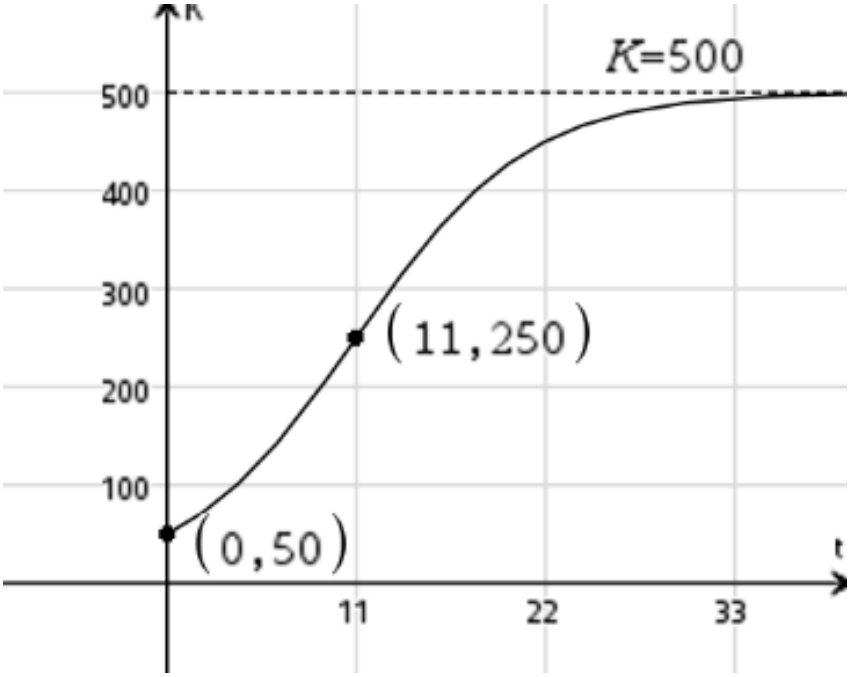
b. $\dfrac{dK}{dt}=\dfrac{500K-K^2}{2500}$, $\dfrac{d^2K}{dt^2}=\left(\dfrac{500-2K}{2500}\right)\dfrac{dK}{dt}=0$ when $K=250$

since the number of kangaroos is increasing most rapidly around 2011,
the inflexion point is at $(11,250)$. The line $K=500$ is a horizontal asymptote.

G1



Question 7

$$S=2\pi\int_a^b y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \text{ with } a=0 \text{ and } b=2$$

$$y=\sqrt{3x+4}=(3x+4)^{\frac{1}{2}},\qquad \frac{dy}{dx}=3\times\frac{1}{2}\times(3x+4)^{-\frac{1}{2}}=\frac{3}{2\sqrt{3x+4}}$$

$$y\sqrt{1+\left(\frac{dy}{dx}\right)^2}=\sqrt{3x+4}\sqrt{1+\frac{9}{4(3x+4)}}$$

A1

$$=\sqrt{3x+4}\sqrt{\frac{(12x+16)+9}{4(3x+4)}}=\frac{1}{2}\sqrt{12x+25}$$

$$S=\pi\int_0^2\sqrt{12x+25}\,dx=\pi\int_0^2(12x+25)^{\frac{1}{2}}\,dx$$ A1

$$S=\pi\left[\frac{2}{3\times12}(12x+25)^{\frac{3}{2}}\right]_0^2$$

M1

$$S=\frac{\pi}{18}\left(49^{\frac{3}{2}}-25^{\frac{3}{2}}\right)=\frac{\pi}{18}(343-125)$$

$$S=\frac{109\pi}{9}$$ A1

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Question 8

To prove $5^{2n}+3n-1$ is divisible by 9 for all $n \in N$.

base case $n=1$ $LHS=5^2+3-1=25+2=27$ this is divisible by 9. A1

The statement is true when $n=1$

Assume it is true when $n=k$ so that $5^{2k}+3k-1$ is divisible by 9, so that we can write $5^{2k}+3k-1=9a$ where $a \in N$. Now consider

$5^{2(k+1)}+3(k+1)-1$

$=5^{2k}\times 5^2+3k+3-1$ M1

$=25\times 5^{2k}+3k+2$

$=25\left(5^{2k}+3k-1\right)-72k+27$

$=25\left(5^{2k}+3k-1\right)+9(3-8k)$ M1

$=25\times 9a+9(3-8k)$

$=9(25a+3-8k)$

$=9b$ where $b \in N$ so it is divisible by 9. A1

Since it is true when $n=1$ and assuming it is true when $n=k$ then it is true when $n=k+1$, so by the principle of mathematical induction, it is true for $n \in N$.

Question 9

a. $\underset{\sim}{r}(s,t)=(3+4t+3s)\underset{\sim}{i}+(-2+t+2s)\underset{\sim}{j}+(1+2t-s)\underset{\sim}{k}$

$\underset{\sim}{r_1}=4\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k} \qquad \underset{\sim}{r_2}=3\underset{\sim}{i}+2\underset{\sim}{j}-\underset{\sim}{k}$

$$\underset{\sim}{r_1}\times\underset{\sim}{r_2}=\begin{vmatrix}\underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k}\\ 4 & 1 & 2\\ 3 & 2 & -1\end{vmatrix}=\underset{\sim}{i}\begin{vmatrix}1 & 2\\ 2 & -1\end{vmatrix}-\underset{\sim}{j}\begin{vmatrix}4 & 2\\ 3 & -1\end{vmatrix}+\underset{\sim}{k}\begin{vmatrix}4 & 1\\ 3 & 2\end{vmatrix}$$ M1

$\underset{\sim}{r_1}\times\underset{\sim}{r_2}=-5\underset{\sim}{i}+10\underset{\sim}{j}+5\underset{\sim}{k}=-5\left(\underset{\sim}{i}-2\underset{\sim}{j}-\underset{\sim}{k}\right)$

Let $\underset{\sim}{n}=\underset{\sim}{i}-2\underset{\sim}{j}-\underset{\sim}{k}$ be the normal to the plane

The equation of the plane using the normal through the point $(3,-2,1)$

$1(x-3)-2(y+2)-1(z-1)=0$ M1

$x-3-2y-4-z+1=0$

$x-2y-z=6$

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b. $\underset{\sim}{r}(t) = 2\underset{\sim}{i} - \underset{\sim}{j} + z_0\underset{\sim}{k} + t\left(\underset{\sim}{i} - 2\underset{\sim}{j} + c\underset{\sim}{k}\right)$ in parametric form

$x = 2 + t,\quad y = -1 - 2t,\quad z = z_0 + ct$ substitute into the plane $x - 2y - z = 6$

$2 + t - 2(-1 - 2t) - (z_0 + ct) = 6$ M1

$2 + t + 2 + 4t - z_0 - ct = 6$

$(5 - c)t = 2 + z_0$

For no solution we require $c = 5,\quad z_0 \neq -2,\quad z_0 \in R \setminus \{-2\}$ A1

c. The line $\dfrac{x-3}{2} = \dfrac{y+1}{b} = z - 2$ has direction

$\underset{\sim}{v} = 2\underset{\sim}{i} + b\underset{\sim}{j} + \underset{\sim}{k},\quad |\underset{\sim}{v}| = \sqrt{4 + b^2 + 1} = \sqrt{5 + b^2}$

the normal to the plane $\underset{\sim}{n} = \underset{\sim}{i} - 2\underset{\sim}{j} - \underset{\sim}{k},\quad |\underset{\sim}{n}| = \sqrt{1 + 4 + 1} = \sqrt{6}$

$\underset{\sim}{v}.\underset{\sim}{n} = 2 - 2b - 1 = 1 - 2b$

$\sin\left(30^0\right) = \cos\left(60^0\right) = \dfrac{1}{2} = \dfrac{\underset{\sim}{v}.\underset{\sim}{n}}{|\underset{\sim}{v}||\underset{\sim}{n}|} = \dfrac{1 - 2b}{\sqrt{6}\sqrt{\left(5 + b^2\right)}}$, so we require $1 - 2b > 0,\quad b < \dfrac{1}{2}$

$\sqrt{6\left(5 + b^2\right)} = 2(1 - 2b)$

$6\left(5 + b^2\right) = 4\left(1 - 4b + 4b^2\right)$

$3\left(5 + b^2\right) = 2\left(1 - 4b + 4b^2\right)$ M1

$15 + 3b^2 = 2 - 8b + 8b^2$

$5b^2 - 8b - 13 = 0$

$(5b - 13)(b + 1) = 0$

$b = \dfrac{13}{5},\quad b = -1$

but since $b < \dfrac{1}{2}$

$b = -1$ only. A1

Question 10

a. $I_0 = \int_0^1 xe^{x^2}\,dx$, let $u = x^2$ $\dfrac{du}{dx} = 2x$ terminals when $x = 1,\ u = 1$ and when $x = 0,\ u = 0$

$I_0 = \dfrac{1}{2}\int_0^1 e^u\,du = \left[\dfrac{1}{2}e^u\right]_0^1$

$I_0 = \dfrac{1}{2}(e - 1)$ A1

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b. $I_n = \int_0^1 x^{2n+1} e^{x^2}\, dx$ using integration by parts

let $u = e^{x^2}$ $\quad \dfrac{dv}{dx} = x^{2n+1}$

$\dfrac{du}{dx} = 2xe^{x^2} \quad v = \dfrac{1}{2n+2} x^{2n+2} = \dfrac{x^{2n+2}}{2(n+1)}$ A1

$$I_n = \left[\frac{x^{2n+2}e^{x^2}}{2(n+1)}\right]_0^1 - \int_0^1 \frac{x^{2n+2}}{2(n+1)} 2xe^{x^2}\, dx$$ M1

$$I_n = \frac{e}{2(n+1)} - \frac{1}{n+1}\int_0^1 x^{2n+3} e^{x^2}\, dx$$

$$I_n = \frac{e}{2(n+1)} - \frac{1}{n+1}\int_0^1 x^{2(n+1)+1} e^{x^2}\, dx$$

$$I_n = \frac{e}{2(n+1)} - \frac{1}{n+1} I_{n+1}$$ M1

$$\frac{1}{n+1} I_{n+1} = \frac{e}{2(n+1)} - I_n$$

$$I_{n+1} = \frac{e}{2} - (n+1) I_n$$

c. when $n = 0$, $I_1 = \dfrac{e}{2} - I_0 = \dfrac{e}{2} - \dfrac{1}{2}(e-1) = \dfrac{1}{2}$

when $n = 1$, $I_2 = \dfrac{e}{2} - 2I_1 = \dfrac{e}{2} - 1$

when $n = 2$, $I_3 = \dfrac{e}{2} - 3I_2 = \dfrac{e}{2} - 3\left(\dfrac{e}{2} - 1\right) = 3 - e$ A1

End of detailed answers for the
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